

Ixmucané GeoPower

Website Editing & Deployment Guide

Prototype repository: github.com/julimo80/Ixmucane-geopower

September 29, 2026

Overview

The site is an Astro + Tailwind static site. Almost all real content — copy, stats, service descriptions, case studies, team bios, photo/video captions — lives in plain JSON files under `src/data/`. You should not need to touch the `.astro` component files (the layout/code) for day-to-day content edits; only for structural changes.

1 Running the Prototype

The project needs Node.js ≥ 22 and ffmpeg (for video work). Both are installed in a dedicated conda environment, `ixmucane-web`, kept separate from your other project environments so nothing else on the machine is affected.

```
# one-time per terminal session
conda activate ixmucane-web

cd ~/Ixmucane-geopower

npm run dev          # dev server at http://localhost:4321, hot-reloads on save
```

Note: a dev server may already be running in the background from an earlier session. If `npm run dev` complains the port is taken, either reuse the one already running (just edit & refresh the browser) or stop it first with `pkill -f "astro dev"`.

Other commands, same environment:

```
npm run build        # production build to dist/ -- also catches JSON/syntax
                     # errors before they reach a live site
npm run preview      # serves that production build locally, for a final
                     # sanity check before deploying
```

To stop the dev server: `Ctrl+C` in its terminal, or `astro dev stop` (the project's own convention, see [AGENTS.md](#)).

2 Where to Edit Content

Everything below is in `src/data/*.json`. Edit the text, save, and the dev server updates the page automatically.

File	Controls
<code>site.json</code>	Site name, tagline, nav links, hero headline/lead, stat strip, contact heading/email, footer note.
<code>pages.json</code>	The four link cards on the Home page (Services, Technology, Case Studies, Team, Contact overview text).

File	Controls
<code>flagship.json</code>	The “AI Favorability Mapping” flagship service card and its map caption.
<code>services.json</code>	The seven remaining numbered services (02–08).
<code>technology.json</code>	Technology page: capability list, the 5-step process pipeline, and the <code>methodology.items[].figcap / description</code> fields – the three figure captions (geological model, permeability cross-sections, ESP/GBOP pump sizing).
<code>caseStudies.json</code>	The three case-study cards. Set <code>"draft": false</code> once a case study is client-confirmed; the yellow draft banner disappears automatically. <code>chartCaption</code> controls the chart caption on LOG 03.
<code>team.json</code>	Team bios and the publications list.
<code>fields.json</code>	The six Guatemalan fields plotted on the favorability map (name, status, note).
<code>gallery.json</code>	Home page photo gallery: 8 entries, each with <code>key</code> (must match a filename in <code>src/assets/field/</code>), <code>caption</code> , and <code>note</code> .
<code>fieldReel.json</code>	Home page video collage: 4 clips, each with <code>key</code> , <code>file</code> (filename in <code>public/videos/</code>), and <code>alt</code> (screen-reader text only, not shown visually).

3 Editing Photos and Videos

3.1 If you edit an existing image and keep the same filename

Yes — the build will use your edited version automatically. Astro re-processes every image from `src/assets/` on each build/dev reload and writes a fresh, content-hashed output file, so there is no manual cache-busting step on the build side. The only thing that can look stale is your *browser’s* own cache of the old picture – do a hard refresh (Ctrl+Shift+R) if you don’t see the change.

3.2 If you rename a file or switch to a different photo

This depends on *how* that image reaches the page:

- **Home page gallery & video collage** (`gallery.json` / `fieldReel.json`): fully key-driven, no code changes needed. Drop the new file into `src/assets/field/` (photos) or `public/videos/` (clips), then update `key` / `file` in the JSON to match the new filename.
- **Everything else** (Hero, Nav logo, Team photos, Technology figures, Case Study chart): these are imported by exact path at the top of the relevant `.astro` component (e.g. `src/components/Team.astro` line 4: `import julianPhoto from '../assets/team/julian-xicara.webp'`). To swap one of these, either **overwrite the existing file in place** (simplest – no code edit), or replace the file and edit that one `import` line to point at the new filename.

Rule of thumb: same filename, no code touch, ever. Different filename: check whether it is JSON-driven (edit JSON only) or component-imported (edit one import line).

3.3 Adding a brand-new photo to the gallery (not just replacing one)

1. Put the file in `src/assets/field/`, e.g. `new-shot.jpg`.

2. Add an entry to `gallery.json`:

```
{ "key": "new-shot", "caption": "...", "note": "..." }
```

3. Save – it appears in the grid automatically (the component globs the folder by key, no import statement needed for gallery photos).

4 Verifying Your Changes

1. Keep `npm run dev` running and check the page in a browser at <http://localhost:4321>.
2. Before considering something final, run `npm run build` once – it will fail loudly on a malformed JSON file or a broken image path, which the dev server sometimes lets slide silently.

5 Before Going Live

5.1 Content still marked as placeholder

The footer currently says outright: *“Prototype for review — case studies, team bios, publications and contact email are placeholders pending confirmation, not live content.”* Concretely, that means:

- **Photo/video captions** – every entry in `gallery.json` and `fieldReel.json` has a [Placeholder] note asking you to confirm the field, well name, and year before publishing.
- **Case studies** – LOG 01 and LOG 02 in `caseStudies.json` are marked `"draft": true` (real content, unconfirmed client). LOG 03 (Tecuamburro–Infiernitos) is real and confirmed.
- **Publications** – `team.json` has one placeholder entry inviting further publications/conference proceedings to be added.
- **Contact email** – `site.json`’s `contact.email` (currently `hello@ixmucanegeopower.com`) needs to be a real, monitored mailbox before launch.
- Once everything above is confirmed, delete or rewrite `site.json`’s `footerNote` – it exists specifically to flag this prototype status to reviewers.

5.2 Deployment (turning this into a live, public website)

The project is a fully static site (`npm run build` produces plain HTML/CSS/JS/images in `dist/` – no server or database needed), so hosting is simple:

1. **Pick a static host** and connect it to the GitHub repository (github.com/julimo80/Ixmucane-geopower). Any of these auto-detect Astro, run `npm run build`, and redeploy on every push to `main`:
 - [Vercel](#)
 - [Netlify](#)
 - [Cloudflare Pages](#)

All three have a free tier sufficient for a marketing site like this one, and all three provision HTTPS/SSL automatically.

2. **Domain name.** If `ixmucanegeopower.com` (or whichever domain you want) is not already registered, register it through any domain registrar. Point its DNS at the host chosen in step 1 – each host’s dashboard gives you the exact records to add.
3. **Contact form (optional upgrade).** The Contact page currently uses a plain `mailto:` link (no backend needed, works immediately). If you would rather have an in-page form, a service

like [Formspree](#) or the hosting platform's own form handling (e.g. Netlify Forms) can be wired in without standing up a custom backend.

4. **Analytics (optional).** None is wired in yet. A privacy-friendly option such as Plausible, or Google Analytics if you prefer, can be added as a small script tag in `src/layouts/Base.astro`.
5. **SEO basics (optional, nice-to-have).** The page already has a title/meta description per page. Not yet present: a [sitemap.xml](#), a real Open Graph share image, and [robots.txt](#) – worth adding once the domain is live, low effort via an Astro integration ([@astrojs/sitemap](#)).

Quick Reference

```
conda activate ixmucane-web          # once per terminal
cd ~/Ixmucane-geopower

npm run dev                          # edit + preview live
npm run build                        # verify a clean production build
npm run preview                      # preview that production build

# edit captions/copy:
src/data/gallery.json               # Home page photo captions
src/data/fieldReel.json            # Home page video alt text
src/data/technology.json           # Technology page figure captions
src/data/caseStudies.json          # Case study text + draft flags
src/data/team.json                 # Team bios + publications
src/data/site.json                 # Site-wide copy, contact email, footer
```